Iterated Symmetric Centipede: Two-Population Learning and Evolutionary Games

2026

Abstract

We study the Iterated Symmetric Centipede (ISC) game under a *two-population* framework. The ISC involves repeated plays of the One-shot Symmetric Centipede (OSC), in which the first-moving player is chosen at random. We consider two separate populations, each using the strategies All-M, All-1, and Grim, with every player from Population 1 playing against every player from Population 2 (no within-population interaction). We first identify the Nash equilibria of the two-population ISC bimatrix game $(C, C^\top)$, which includes both symmetric and *asymmetric* equilibria not reachable by one-population dynamics. We then study large-population evolution via the Two-Population Replicator Dynamics (B.1)–(B.2) and small-population evolution via a two-population Markov chain using Pairwise Proportional Imitation (PPI). We show that the threshold $p^* = 2/3$ governing the stability of the all-defector equilibrium is preserved from the one-population model, and that the two-population framework generically produces asymmetric long-run outcomes in which one population cooperates fully while the other plays Grim.

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1 Introduction

This paper extends the framework of Lazaridis and Kehagias [2] by replacing the single evolving population with *two* populations that play exclusively against each other. The motivation is twofold.

First, one-population replicator dynamics can only converge to *symmetric* Nash equilibria [6]. The ISC possesses non-symmetric NE (e.g. Population 1 playing All-M while Population 2 plays Grim), which are strategically meaningful but inaccessible to one-population dynamics. The two-population framework recovers these.

Second, many real interactions are naturally inter-group rather than intra-group: buyers vs. sellers, incumbents vs. entrants, immune systems vs. pathogens. A two-population model captures this structure directly.

The paper is structured to match [2] section by section. Section 2 reviews the classic Centipede and OSC. Section 3 defines the two-population ISC. Section 4 derives the payoff matrix. Section 5 identifies all Nash equilibria, including the new asymmetric ones. Section 6 presents the two-population replicator dynamics and stability analysis. Section 7 discusses evolutionary stability. Section 8 analyses the two-population Markov chain. Section 9 connects to learning theory. Section 11 concludes.

2 Preliminaries

2.1 Classic Centipede

The Centipede game [1] is an extensive-form game of perfect information on a path of M non-terminal vertices. Player 1 moves at odd turns and Player 2 at even turns. The total payoff at turn t is $c_t = t$. When the active player terminates at turn t, the payoffs are $(p\,c_t,\ (1-p)\,c_t)$ with terminator's share $p \in (\frac{1}{2}, 1]$. We define the strategy family

$$\rho_m = \text{“terminate at the first available turn at or after stage } m\text{”}.$$

Backward induction yields the unique subgame-perfect equilibrium (ρ_1, ρ_1), producing payoffs $(p, 1-p)$. Both players would fare better under (ρ_M, ρ_M), which produces $(pM, (1-p)M)$.

2.2 One-Shot Symmetric Centipede (OSC)

To remove asymmetry we randomise the first mover: each player moves first with probability $\frac{1}{2}$. The resulting normal-form game has $M \times M$ payoff matrix A for the row player:

$$A(i,j) = \begin{cases} i & \text{if } i = j, \\ (2i+1)\,p & \text{if } i < j, \\ (2j+1)\,(1-p) & \text{if } i > j. \end{cases} \tag{1}$$

Four values used throughout: $A_{MM} = M$, $A_{11} = 1$, $A_{M1} = 3(1-p)$, $A_{1M} = 3p$.

3 Two-Population ISC

3.1 Setup

The ISC consists of T repeated OSC plays with full history observation. In the two-population model:

- **Population 1** has N_1 players, each using one strategy from $\{\sigma_1, \sigma_2, \sigma_3\}$.
- **Population 2** has N_2 players, each using one strategy from $\{\sigma_1, \sigma_2, \sigma_3\}$.
- Every Population-1 player plays *exactly once* against every Population-2 player. **No within-population games occur.**

3.2 Strategies

Definition 3.1 (ISC Strategies)**.**

1. *σ_1 (**All-**M): play ρ_M in every round.*
2. *σ_2 (**All-**1): play ρ_1 in every round.*
3. *σ_3 (**Grim**): play ρ_M until the opponent plays ρ_m with $m < M$; then switch permanently to ρ_1.*

3.3 Bimatrix Game

Since all strategies are available to both populations and payoffs depend only on (own strategy, opponent's strategy), the game is symmetric: Population 1's payoff matrix is C (defined below) and Population 2's payoff matrix is $C^\top$. We therefore write the bimatrix game as $(C, C^\top)$.

4 The ISC Payoff Matrix

4.1 Derivation

$C(i, j)$ is the payoff to a strategy-i player in Population 1 facing a strategy-j player from Population 2 over T OSC rounds.

Proposition 4.1 (ISC Payoff Matrix)**.**

$$C = \begin{pmatrix} TM & 3(1-p)\,T & TM \\ 3p\,T & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{pmatrix}. \tag{2}$$

Proof. All-M never plays a sub-M action, so it never triggers Grim and Grim never triggers against it. Entries $C(1, \cdot)$, $C(\cdot, 1)$, and $C(3, 3)$ all equal TM. For $C(2, 3)$: in round 1 All-1 faces Grim's ρ_M, earning $A_{1M} = 3p$; Grim triggers; rounds $2, \ldots, T$ are mutual ρ_1, earning $A_{11} = 1$ each. Similarly $C(3, 2) = A_{M1} + (T-1) = 3(1-p) + (T-1)$. Remaining entries follow directly from Definition 3.1. □

4.2 Numerical Values ($M = 4$, $T = 10$)

Table 1: Payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$. Population 2 payoff matrix is $C^\top$.

	vs All-M	vs All-1	vs Grim
All-M	40	7.5	40
All-1	22.5	10	11.25
Grim	40	9.75	40

Table 2: Payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$.

	vs All-M	vs All-1	vs Grim
All-M	40	12	40
All-1	18	10	10.8
Grim	40	10.2	40

Remark 4.1 (Symmetry of C)**.** *$C(1,3) = C(3,1) = C(1,1) = C(3,3) = TM$ and $C(3,2) = 3(1-p) + (T-1) \approx T$, $C(2,3) = 3p + (T-1) \approx T$ for large T. The game is symmetric: $C^\top(j,i) = C(j,i)$, confirming the bimatrix representation $(C, C^\top)$.*

5 Nash Equilibria

Definition 5.1 (Two-Population Nash Equilibrium)**.** *A pair of mixed strategies $(\hat{x}, \hat{y})$ with $\hat{x}, \hat{y} \in \Delta_3$ is a* Nash equilibrium *of the bimatrix game $(C, C^\top)$ iff*

$$\forall x \in \Delta_3 : \ \hat{x}^\top C \hat{y} \geq x^\top C \hat{y}, \qquad \forall y \in \Delta_3 : \ \hat{y}^\top C^\top \hat{x} \geq y^\top C^\top \hat{x}.$$

Note that $\hat{y}^\top C^\top \hat{x} = \hat{x}^\top C \hat{y}$, so both players share the same equilibrium payoff.

Proposition 5.1 (Asymmetric NE Family)**.** *For every $M \in \{3, 4, \ldots\}$ and large enough T, the following are NE of the two-population ISC for any $a', a'' \in [0,1]$:*

$$\big((1-a',\, 0,\, a'),\ (1-a'',\, 0,\, a'')\big). \tag{3}$$

Proof. Let $\hat{x} = (1-a', 0, a')$ and $\hat{y} = (1-a'', 0, a'')$. Since $C(i,j) = TM$ for all $i, j \in \{1,3\}$, every strategy in the support of $\hat{x}$ or $\hat{y}$ earns TM against the opponent mixture. The mean payoff is TM. A deviation to σ_2 earns at most $\max(C(2,1), C(2,3)) = \max(3pT,\, 3p + (T-1)) < TM$ for $M \geq 2$. Hence no profitable deviation exists for either population, regardless of the values of a' and a''. □

Remark 5.1 (New equilibria vs. one-population case)**.** *Proposition 5.1 includes* asymmetric *NE such as (e_1, e_3) (Pop 1 all All-M, Pop 2 all Grim) and (e_3, e_1). These cannot be expressed as symmetric NE and are therefore inaccessible to one-population dynamics (which can only locate symmetric equilibria).*

Proposition 5.2 (Characterisation of Symmetric NE)**.** *For every $M \in \{3, 4, \ldots\}$ and large enough T, the following are the only symmetric NE $(\hat{x}, \hat{x})$ of the two-population ISC:*

1. $((1-a', 0, a'), (1-a', 0, a'))$ *for all* $a' \in [0, 1]$.
2. $((0, 1, 0), (0, 1, 0))$ *if and only if* $p \geq \frac{2}{3}$.
3. *When* $p \geq \frac{2}{3}$, *the mixed NE*

$$\left(\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)\right)$$

and

$$\left(\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}\right), \left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}\right)\right).$$

These are exactly the symmetric NE from [2], Proposition 4.3.

Proof. For a symmetric NE $(\hat{x}, \hat{x})$, both populations face the same mixture and the problem reduces to the one-population game. The symmetric NE of the bimatrix game $(C, C^\top)$ coincide with the NE of the one-population game with matrix C. The result then follows from [2], Proposition 4.3. □

Example 5.1 ($M = 4$, $T = 10$, $p = 3/4$)**.** *The NE include: all pairs from* (3) *(e.g.* (e_1, e_3), (e_3, e_1), (e_1, e_1), (e_3, e_3), *and every mixed combination), plus the symmetric mixed NE* $\left((\frac{1}{8}, \frac{7}{8}, 0), (\frac{1}{8}, \frac{7}{8}, 0)\right)$ *and* $\left((0, \frac{115}{116}, \frac{1}{116}), (0, \frac{115}{116}, \frac{1}{116})\right)$ *from Proposition 5.2.*

6 Two-Population Replicator Dynamics

6.1 Model

Let $x(t) = (x_1, x_2, x_3)$ and $y(t) = (y_1, y_2, y_3)$ denote the strategy frequency vectors of Population 1 and Population 2 at time t, respectively. Because there are no within-population games, the fitness of strategy m in Population 1 is $(Cy)_m$ and in Population 2 it is $(x^\top C)_m = (C^\top x)_m$. Both populations share the *same* mean payoff scalar $x^\top Cy$.

The two-population replicator equations are:

$$\forall m: \quad \frac{dx_m}{dt} = x_m\big((Cy)_m - x^\top Cy\big), \tag{4}$$

$$\forall m: \quad \frac{dy_m}{dt} = y_m\big((x^\top C)_m - x^\top Cy\big). \tag{5}$$

The system (4)–(5) conserves the simplices $\Delta_3^x = \{x \geq 0, \sum x_m = 1\}$ and $\Delta_3^y = \{y \geq 0, \sum y_m = 1\}$ separately.

Remark 6.1 (Connection to one-population dynamics)**.** *Setting $y = x$ and requiring consistency yields $dx_m/dt = x_m[(Cx)_m - x^\top Cx]$, the standard one-population replicator. Hence the one-population dynamics is the restriction of* (4)–(5) *to the diagonal $x = y$.*

6.2 Rest Points

Proposition 6.1 (Rest Points)**.** *For every $M \in \{3, 4, \ldots\}$ and every T, the following are admissible rest points of* (4)–(5)*:*

1. *For all $a, b \in [0, 1]$:*
$$\big((1-a,\ 0,\ a),\ (1-b,\ 0,\ b)\big). \tag{6}$$
2. $\big((0, 1, 0),\ (0, 1, 0)\big)$ *for all p.*
3. *When $p \geq \frac{2}{3}$, the symmetric mixed rest points from Proposition 5.2.*

The set (6) *is a two-dimensional non-isolated equilibrium set.*

Proof. For (6): if $x_2 = y_2 = 0$ then $(Cy)_m = TM$ for $m \in \{1, 3\}$ (since $C(1, j) = C(3, j) = TM$ for $j \in \{1, 3\}$). Hence $(Cy)_m = x^\top Cy = TM$ and $\dot{x}_m = 0$. Similarly $(x^\top C)_m = TM$ for $m \in \{1, 3\}$ and $\dot{y}_m = 0$. The set is two-dimensional (parametrised by a and b independently).

For $((0, 1, 0), (0, 1, 0))$: $(Cy)_2 = C(2, 2) = T = x^\top Cy$ and $(x^\top C)_2 = C(2, 2) = T$. All derivatives are zero.

Symmetric mixed rest points follow by the same argument as Proposition 4.1 in [2]. □

6.3 Stability of Rest Points

Proposition 6.2 (Stability of the Cooperative Face)**.** *For every $M \geq 4$ and large enough T, the entire face $\mathcal{F} = \{(x, y) : x_2 = y_2 = 0\}$ is a stable non-isolated equilibrium set. Specifically, for any $(x^*, y^*) \in \mathcal{F}$ and perturbations $\varepsilon_x > 0$ (All-1 entering Pop 1) and $\varepsilon_y > 0$ (All-1 entering Pop 2), the Jacobian eigenvalues in the x_2 and y_2 directions are both strictly negative.*

Proof. The eigenvalue governing All-1 invasion of Population 1 is

$$\lambda_{x_2} = (Cy^*)_2 - x^{*\top} Cy^* = \big[b\, C(2, 1) + (1-b)\, C(2, 3)\big] - TM,$$

where $y^* = (1-b, 0, b)$. For large T: $(Cy^*)_2 \approx b \cdot 3pT + (1-b) \cdot T = T[1 + (3p-1)b]$. Hence

$$\lambda_{x_2} \approx T[1 + (3p-1)b] - TM = T[1 - M + (3p-1)b].$$

Since $M \geq 4$ and $(3p-1)b \leq 2$ for $b \leq 1$, $p \leq 1$: $1 - M + (3p-1)b \leq 1 - 4 + 2 = -1 < 0$. By symmetry $\lambda_{y_2} < 0$ for the same reason. Hence $\mathcal{F}$ is stable in both transverse directions. □

Remark 6.2. *Proposition 6.2 shows the cooperative face $\mathcal{F}$ is* always *stable for $M \geq 4$, regardless of p. This contrasts with the one-population model where the corresponding face was only neutrally stable and not asymptotically stable. The key difference is that in the two-population model the All-1 mutant in Population 1 faces only Population 2's cooperation, giving it payoff $\approx T < TM$, while in the one-population model the mutant also encounters the other one-population players at a rate that can sometimes be favourable.*

Proposition 6.3 (Stability of All-1 Equilibrium)**.** *For every $M \in \{3, 4, \ldots\}$ and large enough T:*

1. *If $p > \frac{2}{3}$: $((0, 1, 0), (0, 1, 0))$ is asymptotically stable.*

2. *If* $p \leq \frac{2}{3}$*:* $((0,1,0),(0,1,0))$ *is unstable.*

Proof. The eigenvalues of the Jacobian of (4)–(5) at $((0,1,0),(0,1,0))$ in the directions of each strategy $m \neq 2$ are:

$$\begin{aligned}
\lambda_{x_1} &= C(1,2) - C(2,2) = 3(1-p)T - T = T(2-3p), \\
\lambda_{x_3} &= C(3,2) - C(2,2) = 3(1-p) + (T-1) - T = 2 - 3p, \\
\lambda_{y_1} &= (C^\top e_2)_1 - e_2^\top C e_2 = C(1,2) - T = T(2-3p), \\
\lambda_{y_3} &= C(3,2) - T = 2 - 3p.
\end{aligned}$$

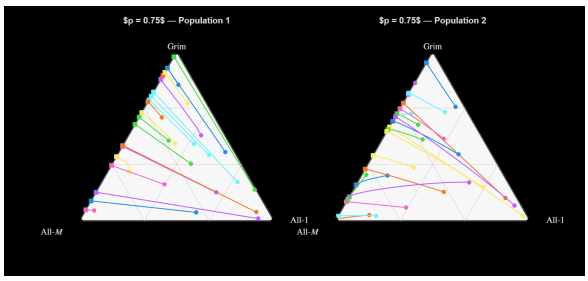
All four eigenvalues have sign $\mathrm{sgn}(2-3p)$. For $p > 2/3$: negative $\Rightarrow$ asymptotically stable. For $p < 2/3$: positive $\Rightarrow$ unstable. For $p = 2/3$: a higher-order perturbation analysis (analogous to Proposition 5.2 in [2]) shows instability. □

Proposition 6.4 (Stability of Asymmetric Pure Equilibria)**.** *For every* $M \geq 4$ *and large enough* T*, the asymmetric pure equilibria* (e_1, e_3) *and* (e_3, e_1) *are* neutrally stable*: the Jacobian eigenvalues are all non-positive, with zero eigenvalues in the directions* $(e_1 - e_3)^x$ *and* $(e_1 - e_3)^y$ *corresponding to the neutrality within the cooperative face.*

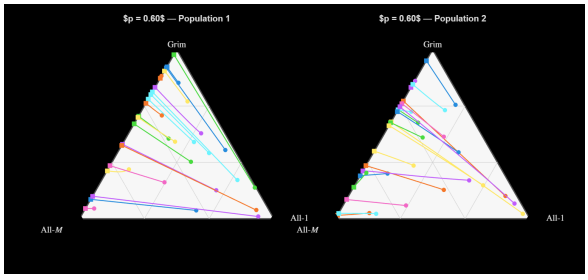
Proof. At (e_1, e_3): $(Ce_3)_m = C(m,3)$, which equals TM for $m \in \{1,3\}$ and $C(2,3) \approx T < TM$ for $m = 2$. Eigenvalues: $\lambda_{x_1} = C(1,3) - TM = 0$, $\lambda_{x_2} = C(2,3) - TM < 0$, $\lambda_{x_3} = C(3,3) - TM = 0$. $(e_1^\top C)_n = C(1,n)$: $\lambda_{y_1} = C(1,1) - TM = 0$, $\lambda_{y_2} = C(1,2) - TM < 0$, $\lambda_{y_3} = C(1,3) - TM = 0$. All eigenvalues are ≤ 0; the zeros reflect neutral drift within $\mathcal{F}$. □

6.4 Phase Portraits

We visualise the dynamics on two side-by-side ternary simplices: the left panel shows Population 1's trajectory and the right panel shows Population 2's trajectory for the same integrated orbit. Both populations start from the same initial condition and are integrated jointly under (4)–(5).

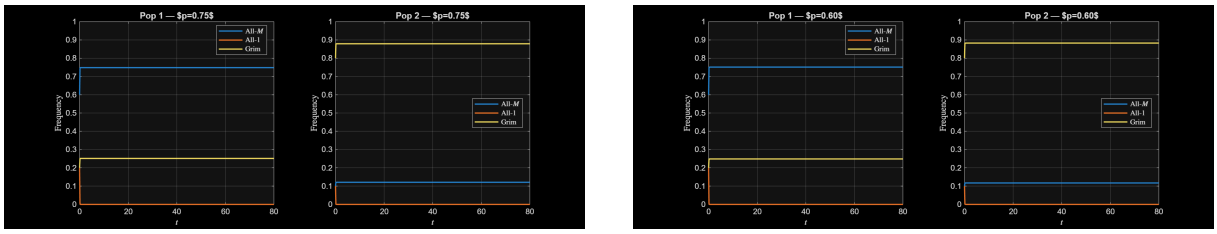


(a) $p = 3/4$: both populations converge to the cooperative face or the All-1 vertex.



(b) $p = 3/5$: All-1 is unstable; both converge to the cooperative face, possibly at different points (asymmetric equilibrium).

Figure 1: Phase portraits of (4)–(5), $M = 4$, $T = 10$. *Left*: Population 1 simplex (vertices: All-M bottom-left, All-1 bottom-right, Grim top). *Right*: Population 2 simplex (same layout). Circle = initial condition; square = final state.



(a) $p = 3/4$. (b) $p = 3/5$.

Figure 2: Time series of (4)–(5), $M = 4$, $T = 10$. Solid lines: Population 1. Dashed lines: Population 2. Colours: blue = All-M, red = All-1, green = Grim. Initial condition: $x_0 = (0.6, 0.2, 0.2)$, $y_0 = (0.1, 0.1, 0.8)$.

6.5 Parameter Sensitivity

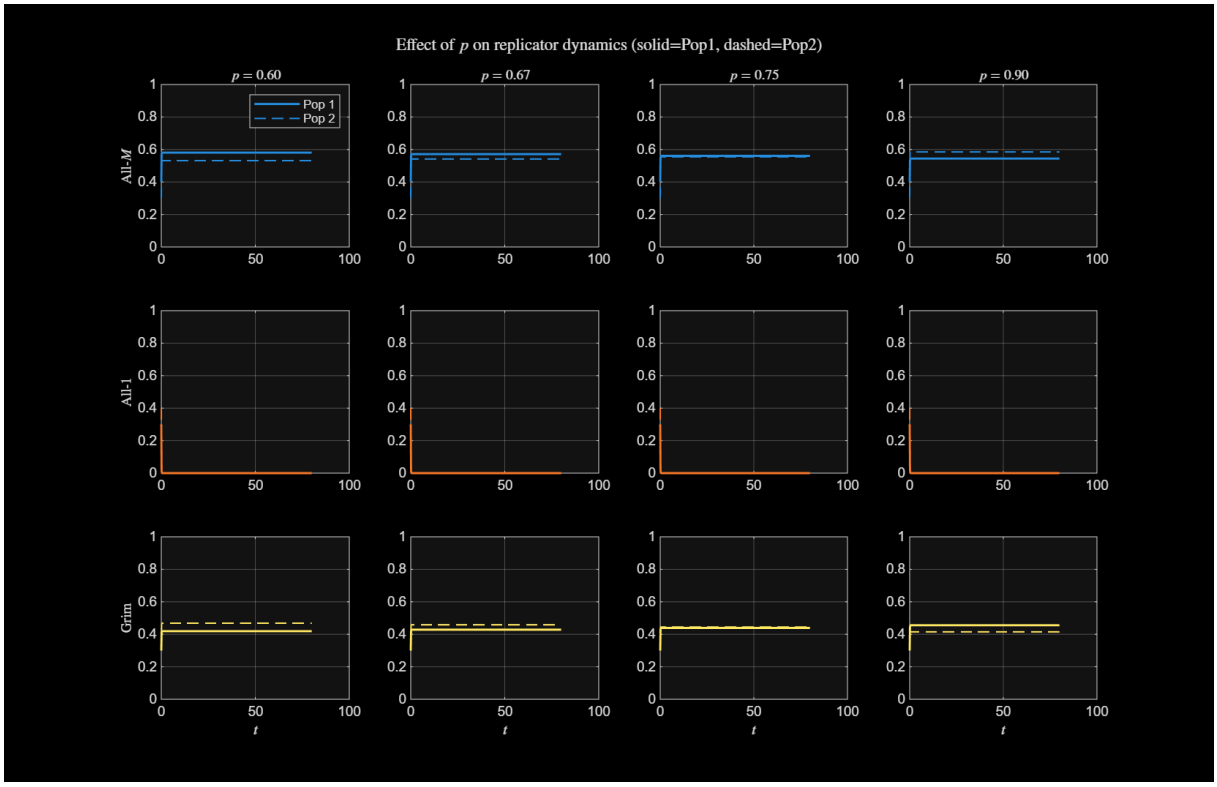


Figure 3: Effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$). Rows: strategy frequencies. Columns: p values. Solid = Pop 1, dashed = Pop 2.

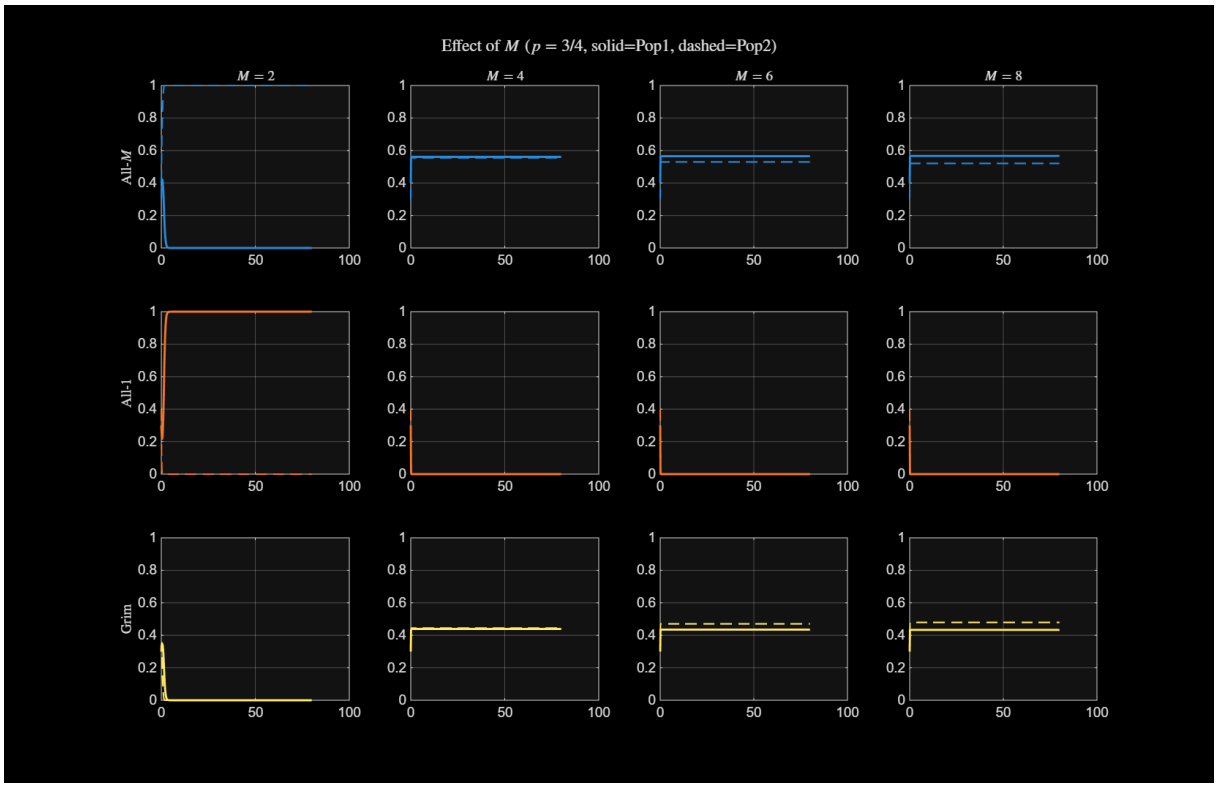


Figure 4: Effect of $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$).

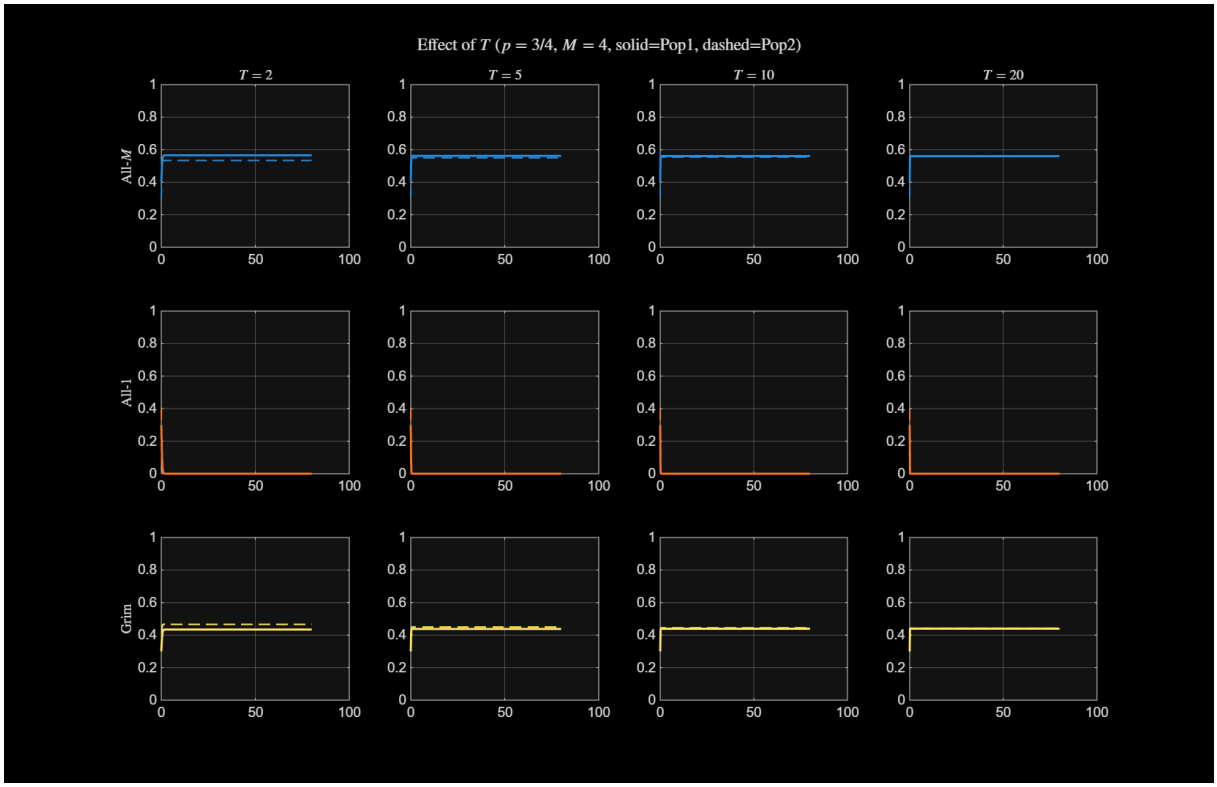


Figure 5: Effect of $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$).

6.6 Initial Condition Sensitivity

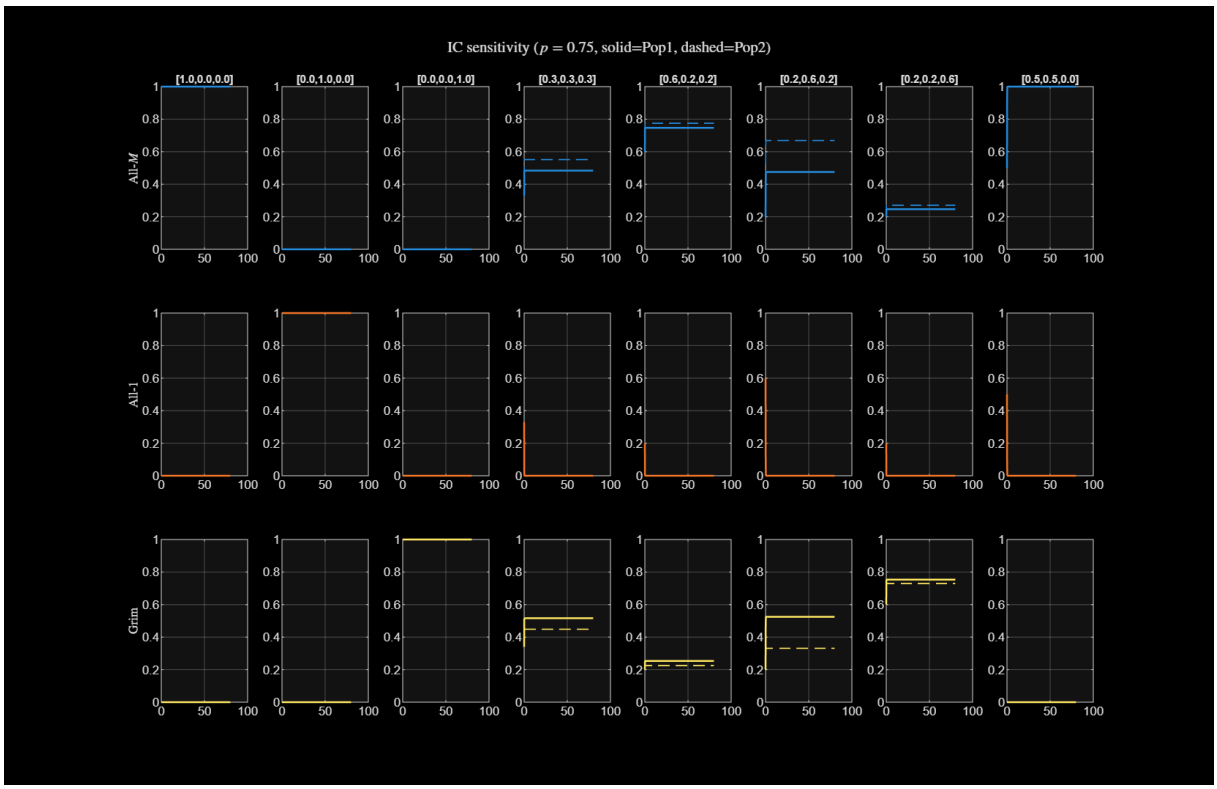


Figure 6: Effect of varying initial condition ($p = 3/4$, $M = 4$, $T = 10$). Symmetric start: $x_0 = y_0$.

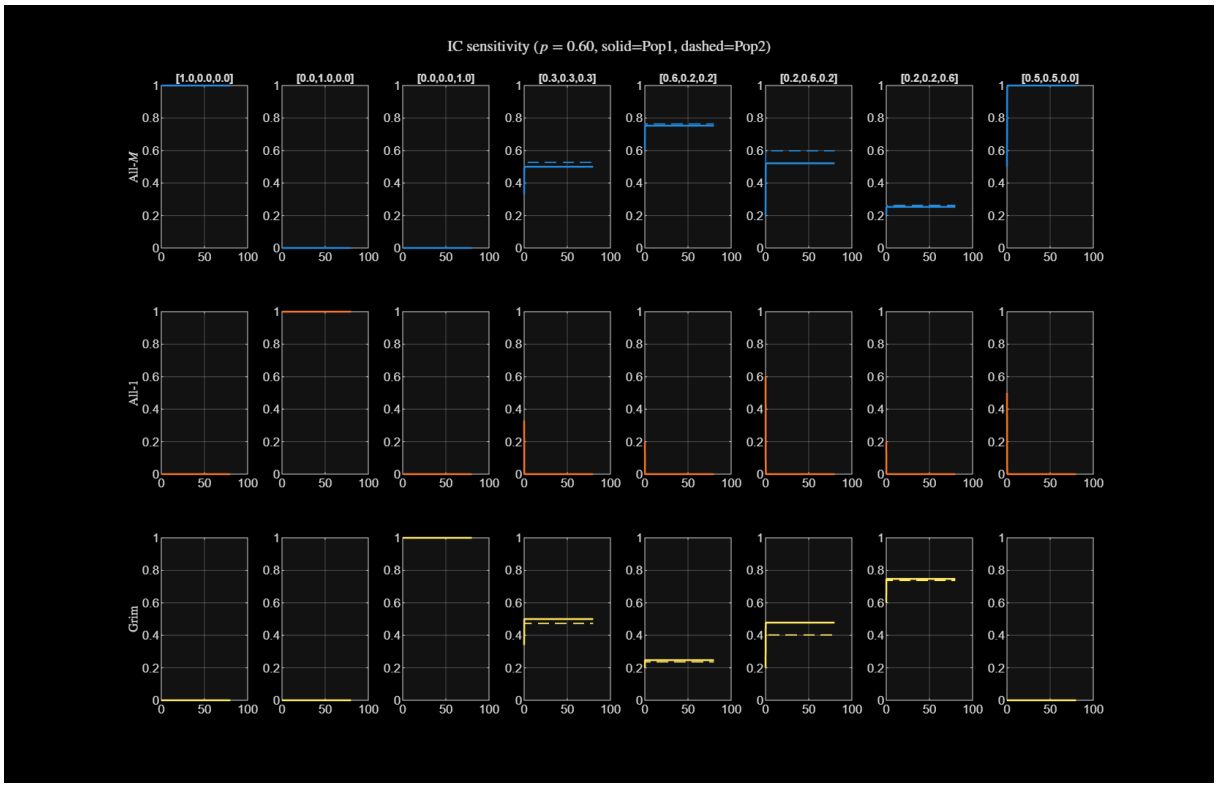


Figure 7: Effect of varying initial condition ($p = 3/5$, $M = 4$, $T = 10$).

7 Evolutionary Stable Strategies

Definition 7.1 (Two-Population ESS)**.** *A strategy $\hat{x} \in \Delta_3$ for Population 1 is* evolutionarily stable *against Population 2 playing $\hat{y} \in \Delta_3$ iff for all $x \neq \hat{x}$:*

$$\hat{x}^\top C\hat{y} \geq x^\top C\hat{y}, \qquad \hat{x}^\top C\hat{y} = x^\top C\hat{y} \;\Rightarrow\; \hat{x}^\top Cy > x^\top Cy.$$

A two-population ESS pair $(\hat{x}, \hat{y})$ *requires this condition for both populations simultaneously.*

Proposition 7.1 (ESS Pairs)**.** *For $M \geq 4$ and large T:*

1. *The pair $((0,1,0),(0,1,0))$ is a two-population ESS iff $p > 2/3$.*
2. *No pure pair (e_i, e_j) with $i, j \in \{1,3\}$ is a two-population ESS: the zero eigenvalues in Proposition 6.4 prevent strict satisfaction of the ESS second condition.*
3. *No interior point of $\mathcal{F}$ is an ESS.*

Proof. Part 1: $\hat{x}^\top C\hat{y} = T$. For $x \neq e_2$: $x^\top Ce_2 = \sum_m x_m C(m,2)$. $C(1,2) = 3(1-p)T < T$ and $C(3,2) = 2 - 3p + T < T$ both hold iff $p > 2/3$. Hence the condition holds strictly, confirming ESS.

Part 2: At (e_1, e_3), $\hat{x}^\top C\hat{y} = C(1,3) = TM$ and $e_3^\top Ce_3 = TM = \hat{x}^\top Ce_3$. The second ESS condition requires $\hat{x}^\top Ce_3 > e_3^\top Ce_3$, i.e. $TM > TM$: false. Hence not ESS.

Part 3: Points in the interior of $\mathcal{F}$ are neutral to perturbations within $\mathcal{F}$ (all earn TM), failing the strict ESS condition. □

8 Markovian Analysis

8.1 Two-Population Markov Chain

The population state is $(\mathbf{s}, \mathbf{t})$ where $\mathbf{s} = (s_1, s_2, s_3) \in \mathbb{Z}^3_{\geq 0}$ with $\sum_m s_m = N_1$ and $\mathbf{t} = (t_1, t_2, t_3) \in \mathbb{Z}^3_{\geq 0}$ with $\sum_n t_n = N_2$. The state space has $\binom{N_1+2}{2} \cdot \binom{N_2+2}{2}$ elements.

Since there are no within-population games, per-player payoffs are:

$$q_m^{(1)}(\mathbf{s}, \mathbf{t}) = \sum_n t_n \, C(m, n), \tag{7}$$

$$q_n^{(2)}(\mathbf{s}, \mathbf{t}) = \sum_m s_m \, C(m, n). \tag{8}$$

Note that (7) is the payoff to a Population-1 strategy-m player against all N_2 Population-2 players, and (8) is the payoff to a Population-2 strategy-n player against all N_1 Population-1 players.

Definition 8.1 (Two-Population PPI)**.** *At each step, a population is chosen uniformly at random. Within the chosen population, one player is selected uniformly and applies the PPI revision protocol:*

$$\varrho^{(1)}_{m_1 m_2}(\mathbf{s}, \mathbf{t}) = \frac{x^{(1)}_{m_2} \left[q^{(1)}_{m_2} - q^{(1)}_{m_1}\right]_+}{\sum_m x^{(1)}_m \left[q^{(1)}_m - q^{(1)}_{m_1}\right]_+},$$

where $x^{(1)}_m = s_m/N_1$. An analogous formula applies within Population 2. The state transition $(\mathbf{s}, \mathbf{t}) \to (\mathbf{s} - \mathbf{e}_{m_1} + \mathbf{e}_{m_2}, \mathbf{t})$ has probability $\frac{1}{2} \cdot (s_{m_1}/N_1) \cdot \varrho^{(1)}_{m_1 m_2}$.

8.2 Absorbing States

Proposition 8.1 (Absorbing States)**.** *For $M \geq 4$, $N_1, N_2 \geq 2$, and large T, the absorbing states of the two-population Markov chain are:*

1. *All states $(\mathbf{s}, \mathbf{t})$ with $s_2 = t_2 = 0$:*

$$\mathcal{A} = \{(\mathbf{s}, \mathbf{t}) : s_1 + s_3 = N_1, \ t_1 + t_3 = N_2, \ s_2 = t_2 = 0\}.$$

   This face has $\binom{N_1+1}{1} \cdot \binom{N_2+1}{1} = (N_1 + 1)(N_2 + 1)$ states.

2. *The pure All-1 state $((0, N_1, 0), (0, N_2, 0))$.*

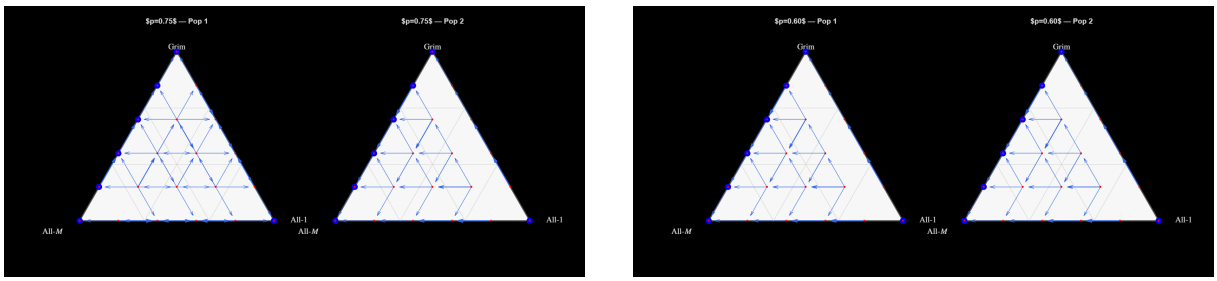
Proof. $\mathcal{A}$ is absorbing. For $(\mathbf{s}, \mathbf{t}) \in \mathcal{A}$ ($s_2 = t_2 = 0$): all Population-1 strategies earn $q^{(1)}_m = \sum_n t_n C(m, n) = TM$ (since $t_2 = 0$ and $C(m, j) = TM$ for $j \in \{1, 3\}$ and $m \in \{1, 3\}$). So all Population-1 players have equal payoff and PPI gives no switches. Similarly for Population 2. Hence $\mathcal{A}$ is absorbing.

$((0, N_1, 0), (0, N_2, 0))$ *is absorbing.* With all players using All-1, there is no other strategy to imitate.

All other states are transient. Any state with $s_2 > 0$ and some $t_m > 0$ for $m \in \{1, 3\}$ has Population-2 strategies earning $q^{(2)}_m = s_2 C(2, m) + (N_1 - s_2)TM$, while Population-2's All-1 earns $q^{(2)}_2 = s_2 C(2, 2) + (N_1 - s_2)C(1, 2)$. For large T: $q^{(2)}_m \approx (N_1 - s_2)TM > q^{(2)}_2 \approx N_1 T$. So All-1 in Population 2 loses fitness to cooperators in Population 2, and with positive probability Population 2's All-1 count decreases. By induction s_2 eventually reaches 0. □

Remark 8.1 (Contrast with one-population case)**.** *In the one-population ISC with All-M, the $s_2 = 0$ face is absorbing because All-M and Grim earn identical payoffs (TM) against each other. In the two-population model this remains true* within *each population's internal comparisons (since $q_m^{(1)} = TM$ for all $m \in \{1, 3\}$ when $t_2 = 0$), so the absorbing face $\mathcal{A}$ is preserved in its full $(N_1 + 1)(N_2 + 1)$-state extent. This is a* larger *absorbing set than the one-population case: the two populations independently choose their level of cooperation along the All-M–Grim spectrum.*

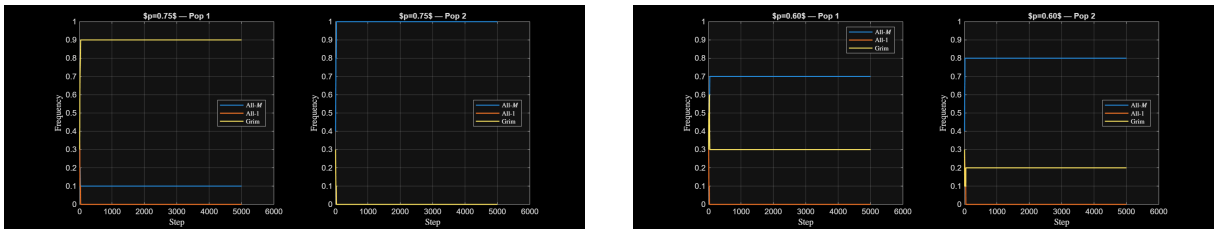
8.3 State Transition Graphs (Figures 4–5)



(a) $p = 3/4$. (b) $p = 3/5$.

Figure 8: Two-population Markov state transition graphs, $M = 4$, $T = 10$, $N_1 = N_2 = 5$. *Left of each subfigure*: Population-1 simplex projection. *Right*: Population-2 simplex projection. Blue dots: states on absorbing face $\mathcal{A}$ or pure All-1. Red dots: transient states. Arrow width $\propto$ transition probability.

8.4 Time Series



(a) $p = 3/4$. (b) $p = 3/5$.

Figure 9: Two-population Markov dynamics time series, $M = 4$, $T = 10$, $N_1 = N_2 = 10$, 5000 steps, $\mathbf{s}_0 = (3, 4, 3)$, $\mathbf{t}_0 = (4, 3, 3)$. Solid: Pop 1. Dashed: Pop 2.

8.5 Parameter Sensitivity

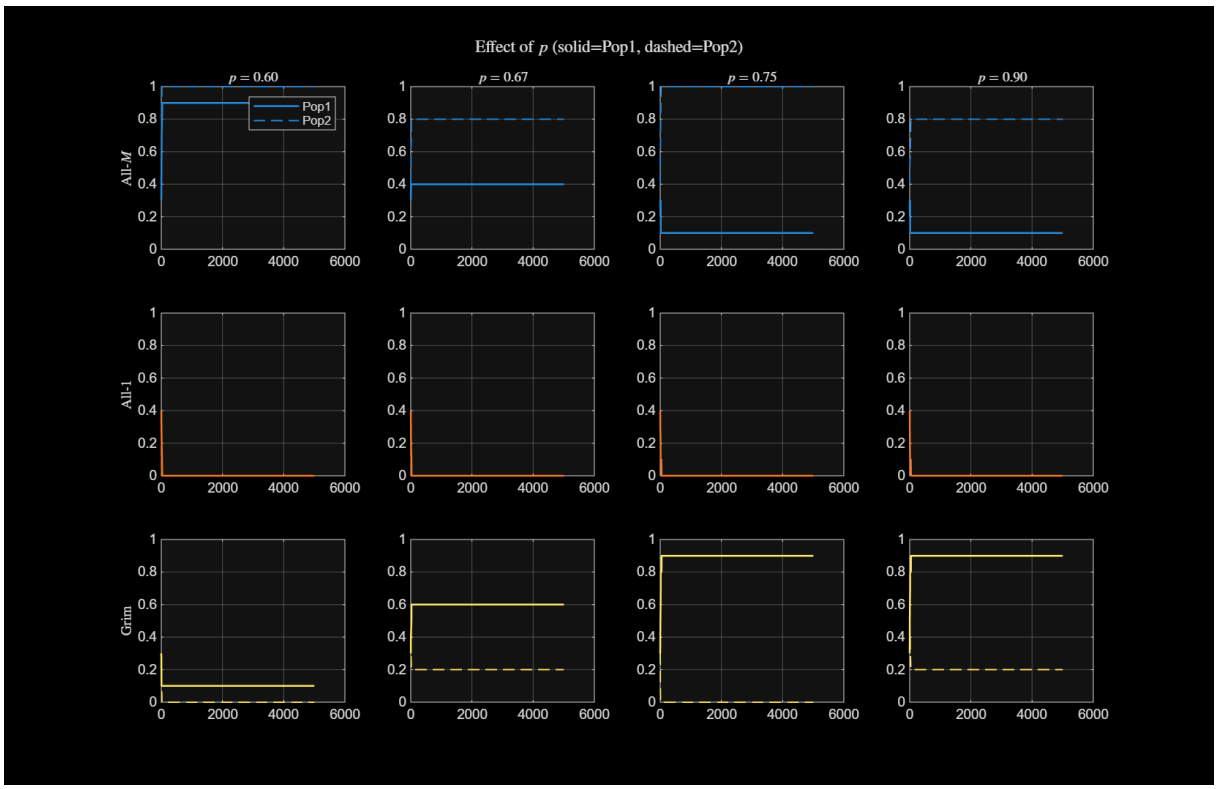


Figure 10: Effect of $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$, $N = 10$).

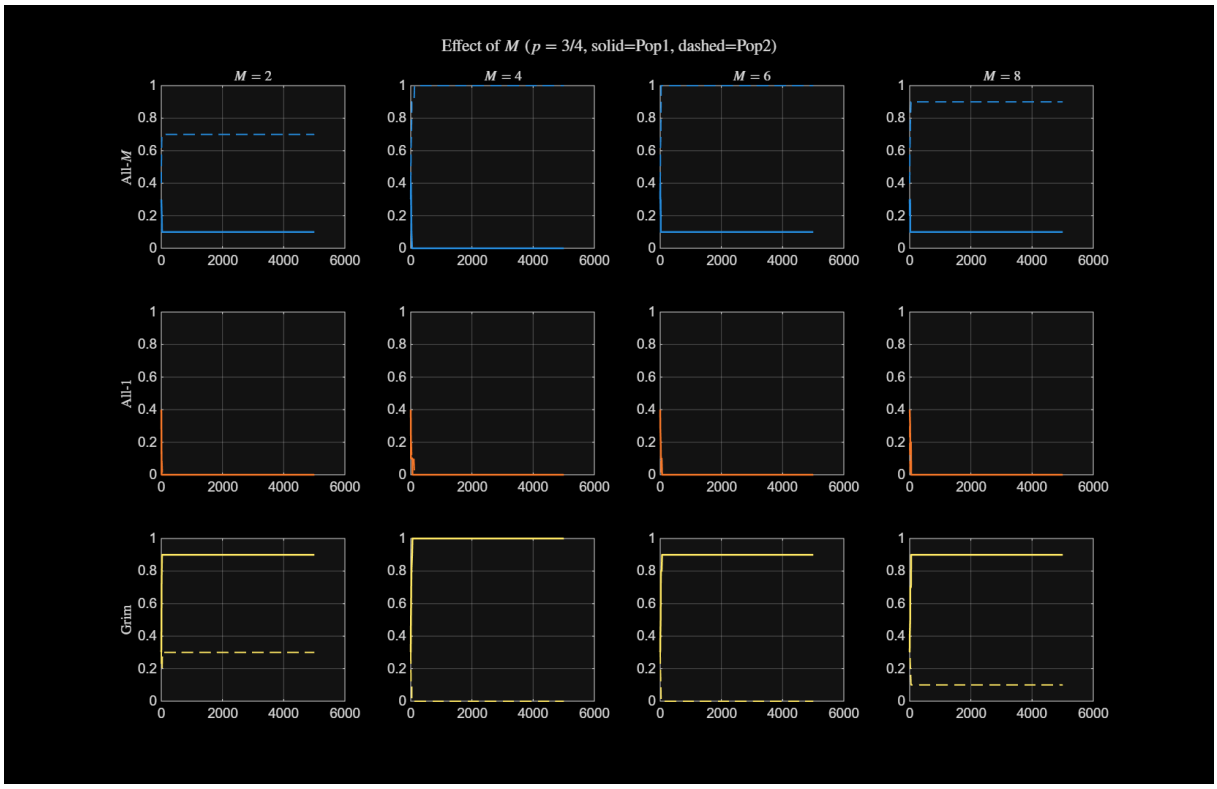


Figure 11: Effect of $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$, $N = 10$).

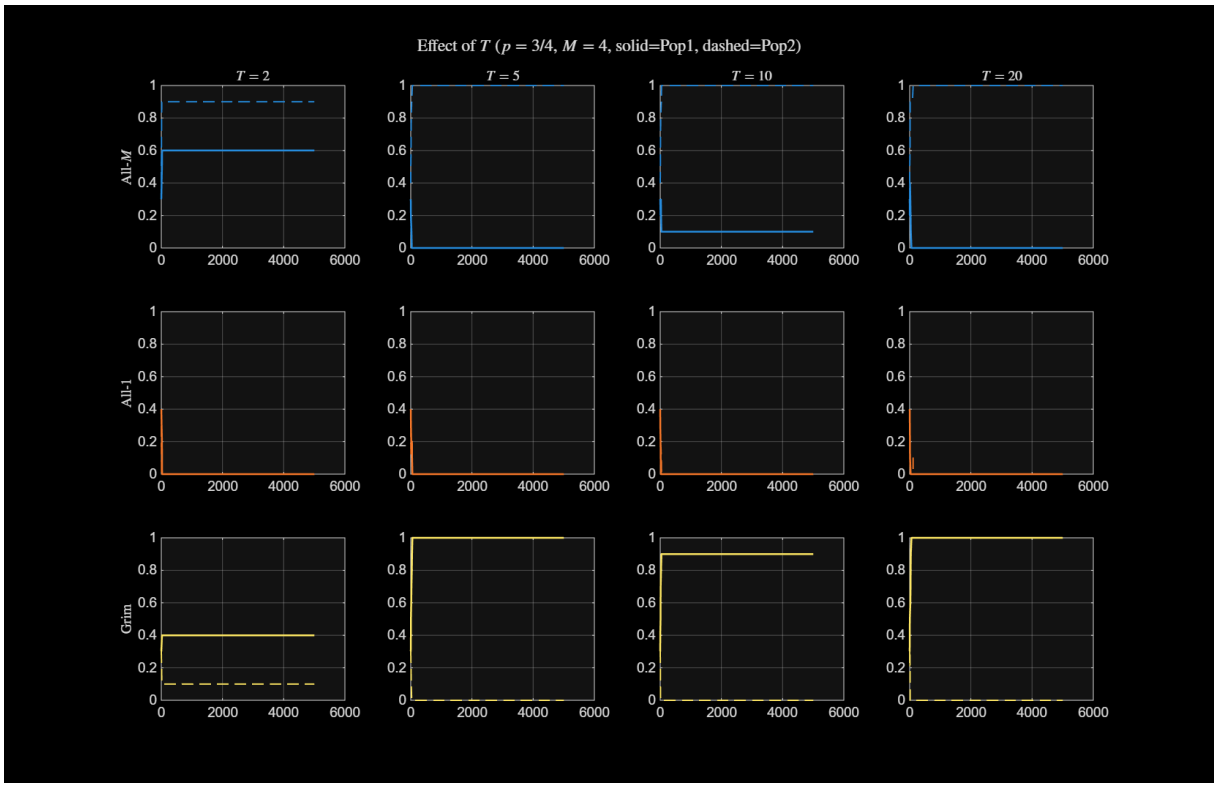


Figure 12: Effect of $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$, $N = 10$).

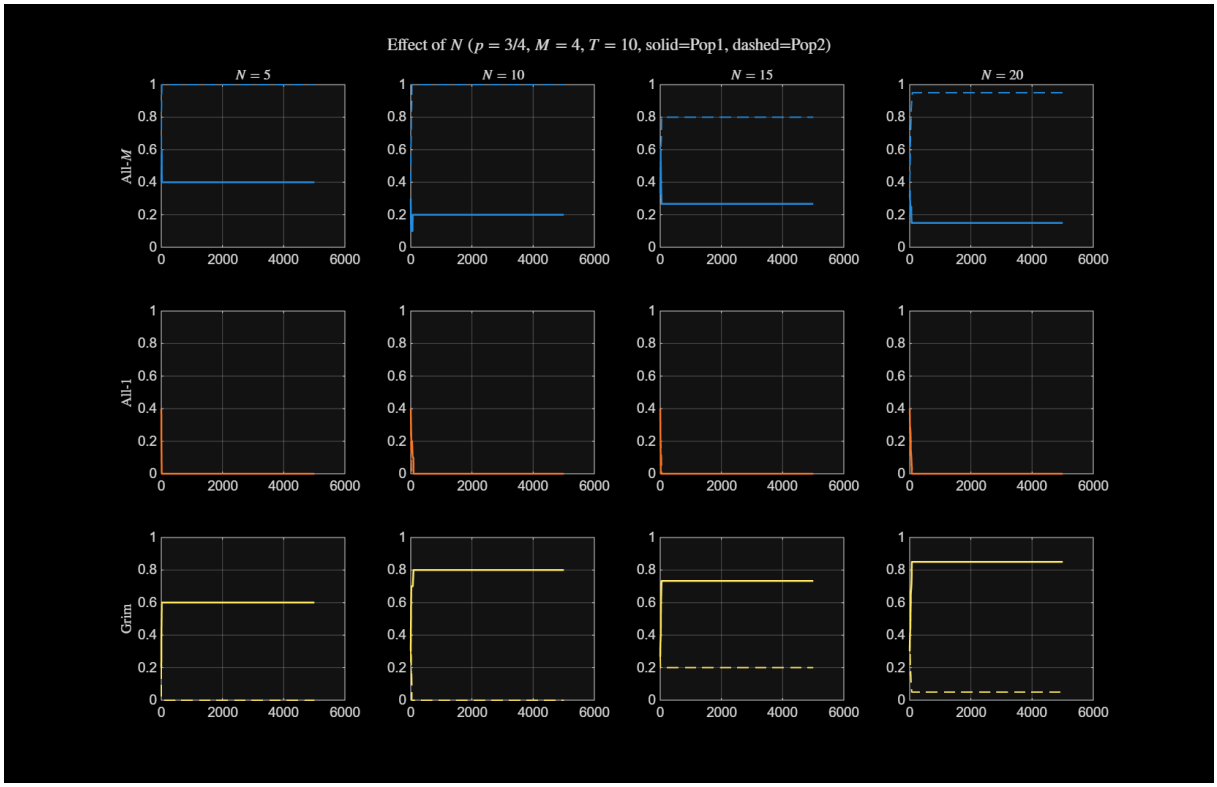


Figure 13: Effect of $N \in \{5, 10, 15, 20\}$ ($N_1 = N_2 = N$, $p = 3/4$, $M = 4$, $T = 10$).

8.6 Initial State Sensitivity

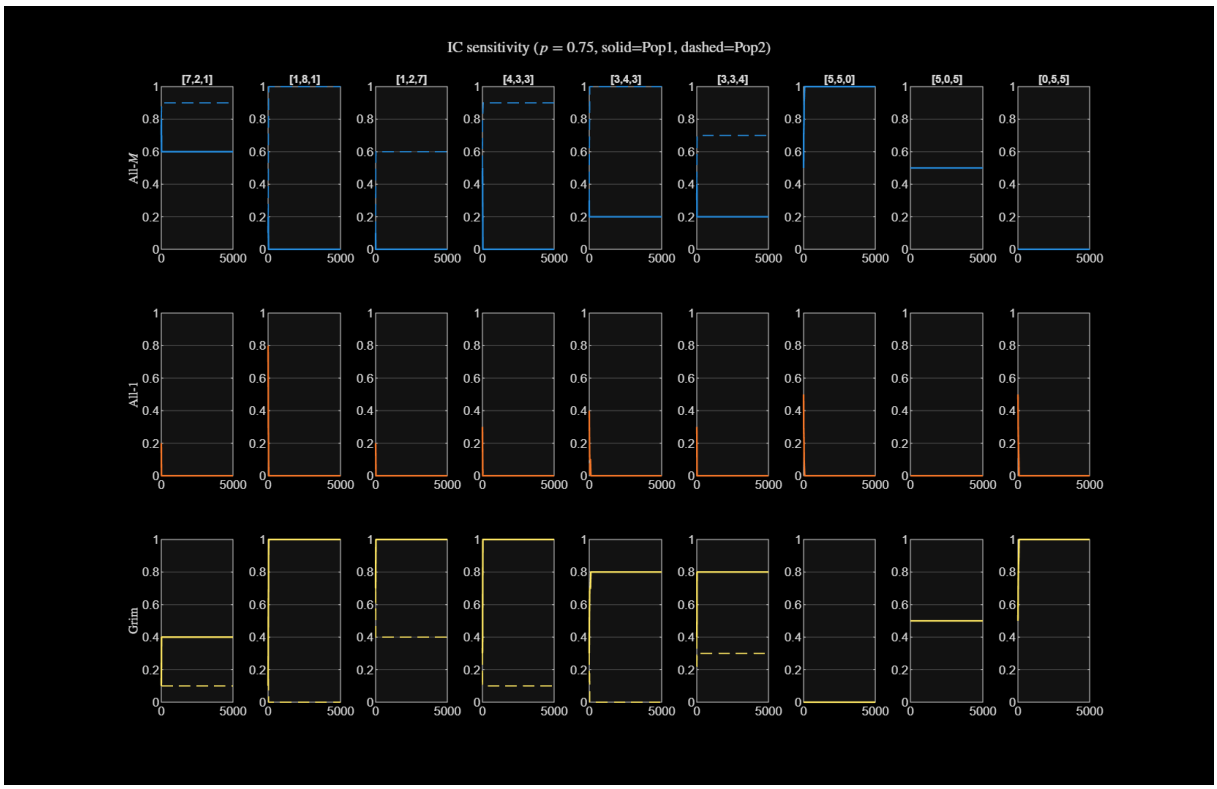


Figure 14: Initial state sensitivity ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$).

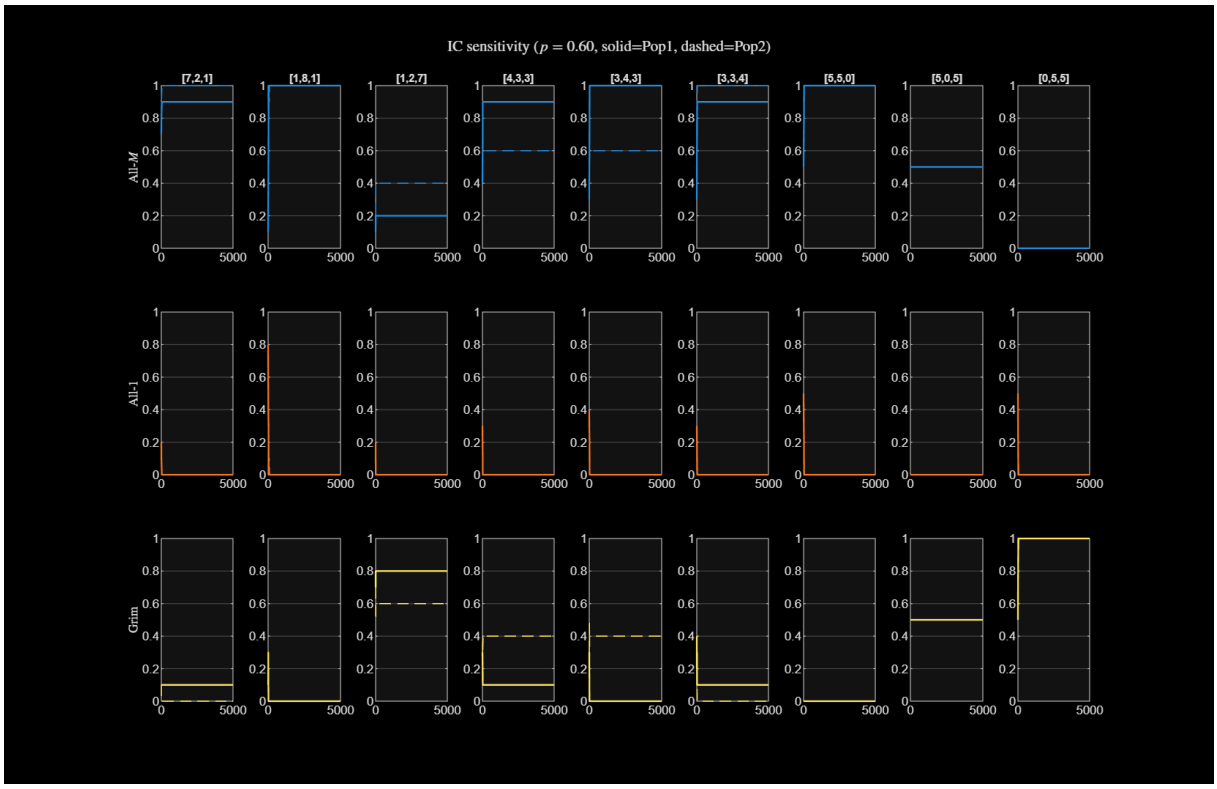


Figure 15: Initial state sensitivity ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$).

9 Connection to Learning Theory

Both evolutionary game theory and individual learning models converge to the same formal dynamics under different interpretations [4].

In the two-population context:

- **Evolutionary interpretation**: Population 1 evolves against Population 2. The cooperative face $\mathcal{F}$ is stable because any All-1 mutant in either population faces only cooperative opponents from the other population, earning $\approx T \ll TM$.
- **Learning interpretation**: Individual agents in Population 1 update their strategies based on payoff comparisons with other Population-1 agents; their payoffs come from Population-2 opponents. This is the natural model for *between-group learning*: players learn from their own group but earn from interactions with the other.
- **PPI and reinforcement**: Under PPI [6], the two-population Markov chain converges in distribution (as $N_1, N_2 \to \infty$) to the replicator dynamics (4)–(5), by an analogous argument to Proposition 7.4 of [2].

10 Discussion

10.1 New Phenomena vs. One-Population Model

Table 3 summarises the key differences between the one-population and two-population ISC models.

Table 3: Comparison: one-population vs. two-population ISC.

Feature	One-population	Two-population
Replicator dynamics	$\dot{x}_m = x_m[(Cx)_m - x^\top Cx]$	Eqs. (4)–(5)
Accessible NE	Symmetric only	Symmetric and asymmetric
Absorbing face	1-D ($s_2 = 0$ line)	2-D ($s_2 = t_2 = 0$ face)
Total absorbing states	$N + 1$	$(N_1 + 1)(N_2 + 1)$
Payoff in $\mathcal{A}$	All-M earns TM, Grim TM	Same
Stability of $\mathcal{F}$	Neutral (eigenvalue = 0)	Asymptotically stable
p^* for All-1 stability	2/3	2/3 (preserved)

10.2 Asymmetric Equilibria and Their Interpretation

The most striking new feature is the reachability of asymmetric equilibria such as (e_1, e_3) (Pop 1 all-cooperating, Pop 2 all-Grim) from generic initial conditions. Under (4)–(5), if Pop 2 evolves faster toward Grim, Pop 1's All-M and Grim both earn TM against Pop 2's Grim, so Pop 1 has no selection pressure to switch. The system can lock into the asymmetric state.

10.3 The $p = 2/3$ Threshold

The critical threshold $p^* = 2/3$ is entirely preserved:

- $p > 2/3$: the All-1 equilibrium is a two-population ESS.
- $p < 2/3$: All-1 is unstable in both populations simultaneously; the system converges to the cooperative face $\mathcal{F}$.

11 Conclusion

We have studied the two-population ISC with strategies All-M, All-1, and Grim. The main results are:

1. **Payoff matrix**: C is the same 3×3 matrix as in the one-population model; the bimatrix game is $(C, C^\top)$.
2. **Nash equilibria**: A continuum of asymmetric NE of the form $((1-a', 0, a'), (1-a'', 0, a''))$ exists for all $a', a'' \in [0, 1]$, in addition to all symmetric NE of the original paper.
3. **Two-population replicator dynamics**: Equations (4)–(5) preserve the $p^* = 2/3$ stability threshold for the All-1 equilibrium. The cooperative face $\mathcal{F}$ is asymptotically stable in the transverse directions (stronger than the one-population result).
4. **ESS**: All-1 is the unique two-population ESS, iff $p > 2/3$.
5. **Markov chain**: The absorbing face $\mathcal{A}$ is two-dimensional, with $(N_1 + 1)(N_2 + 1)$ states, reflecting the independent cooperation levels of the two populations.
6. **Parameter sensitivity**: Results are robust to variations in p, M, T, N, and initial conditions.

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A Two-Population Replicator Dynamics: Explicit Equations

Writing out (4)–(5) explicitly with $\bar{\pi} = x^\top C y$:

$$\begin{aligned}
\dot{x}_1 &= x_1\big[TM\,y_1 + 3(1-p)T\,y_2 + TM\,y_3 - \bar{\pi}\big],\\
\dot{x}_2 &= x_2\big[3pT\,y_1 + T\,y_2 + (3p+T-1)\,y_3 - \bar{\pi}\big],\\
\dot{x}_3 &= x_3\big[TM\,y_1 + (3(1-p)+T-1)\,y_2 + TM\,y_3 - \bar{\pi}\big],
\end{aligned}$$

$$\begin{aligned}
\dot{y}_1 &= y_1\big[TM\,x_1 + 3pT\,x_2 + TM\,x_3 - \bar{\pi}\big],\\
\dot{y}_2 &= y_2\big[3(1-p)T\,x_1 + T\,x_2 + (3(1-p)+T-1)\,x_3 - \bar{\pi}\big],\\
\dot{y}_3 &= y_3\big[TM\,x_1 + (3p+T-1)\,x_2 + TM\,x_3 - \bar{\pi}\big].
\end{aligned}$$

These are implemented in `repDyn2pop.m`.

B Large-T Approximation of C

$$\frac{C}{T} \approx \begin{pmatrix} M & 3(1-p) & M \\ 3p & 1 & 1 \\ M & 1 & M \end{pmatrix}.$$

At large T the off-diagonal entries $C(2,3)/T \approx 1$ and $C(3,2)/T \approx 1$ are both approximately 1, so All-1 earns the same against Grim as against itself. The cooperative strategies both earn M against each other, $M \gg 1$ for $M \geq 4$, explaining why the cooperative face is strongly attracting.

C MATLAB Implementation Notes

All figures are generated by two main scripts:

- `xRep2pop001.m`: integrates (4)–(5) via `ode45` (implemented in `repDyn2pop.m`) and saves all replicator dynamics figures.
- `xMark2pop001.m`: runs the two-population Markov simulations via `sMarkDyn2pop.m` and builds the exact transition matrix via `buildMarkovP2pop.m`.

Supporting files:

- `computeC.m`: computes the 3×3 ISC payoff matrix.
- `repDyn2pop.m`: ODE RHS for equations (4)–(5).
- `xRepDyn2pop.m`: wrapper calling `ode45`.
- `PhasePlot2pop.m`: two-simplex phase portrait.
- `ternary2cart.m`, `drawSimplex.m`: ternary coordinate utilities.

- `sMarkDyn2pop.m`: Monte Carlo Markov simulation.
- `buildMarkovP2pop.m`: exact transition matrix.
- `PStateTransitionGraph2pop.m`: state transition graph.